POKHARA UNIVERSITY

Level: Bachelor | Semester: Fall | Year : 2020
Programme: B E | Full Marks: 100
Course: Artificial Intelligence | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) What is Artificial Intelligence? Discuss the impact of artificial intelligence on society. 7
   b) Define Intelligent Agent .Describe different types of intelligent Agents with an example. 8
2. a) What is an 8-puzzle problem? How can you solve a problem using production system? Explain with example. 7
   b) How is depth limit search efficient with respect to depth first search? Explain with Example. 8
3. a) "A* search is the combination of best first search & greedy search approach". Explain the statement with suitable example. 7
   b) Assume the following facts: 8
      i) Ravi likes all king of food.
      ii) Apples and chicken are food
      iii) Anything anyone eats and is not killed is food
      iv) Ajay eats peanuts and is still alive
      v) Rita eats everything that Ajay eats

   Prove that **Ravi Likes peanuts** using resolution.
4. a) What is a semantic net? Represent the following sentences using semantic network. 8
   - Birds are animals.
   - Birds can fly and lay eggs
   - Robin is a bird
   - Ruby is a Robin
   - Ruby owns a nest

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b) In a survey, One percent of women over 50 have breast cancer. Ninety percent of women who have breast cancer test positive on mammograms. Eight percent of women will have false positives. From the above statistics, apply Bayes theorem to find the probability that a woman has cancer if she has a positive mammogram result. 7

5. a) Write down the algorithm for K- means clustering. Explain K- means clustering with a suitable example. 8

b) What is fuzzy logic? Explain with its applications. 7

6. a) Describe applications of an Expert system? Explain the architecture of Expert System. 8

b) Explain the steps involved in Natural Language Processing with examples. 7

7. Write short notes on: (Any two) 2×

a) Artificial Neural Network

b) Supervised VS Unsupervised

c) Breadth first search

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